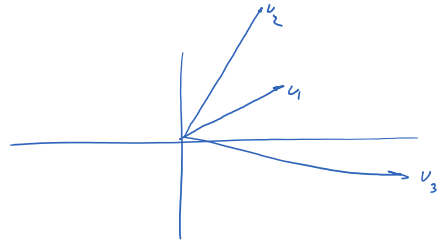
Some Definitions

Def Linear independence

A set of vectors $v_1, v_2, \dots, v_n$ is linearly independent if the linear combination

$$a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$$

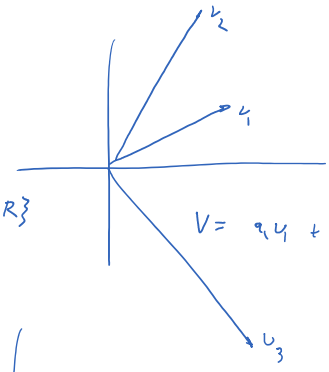
only when $a_1 = a_2 = \dots = a_n = 0$.



$v_3 = a_2 v_2 + a_1 v_1$

Def: span

$$\text{span}((v_1, v_2 \dots v_n)) = \{ v \in S : v = a_1 v_1 + \dots + a_n v_n \text{ for some } a_1, a_2, \dots, a_n \in \mathbb{R} \}$$

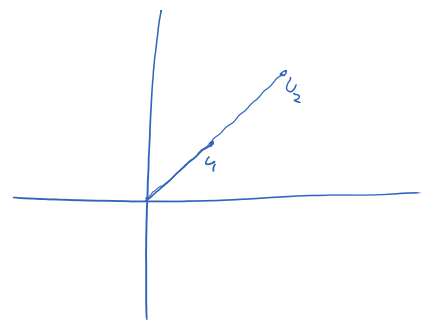


$V = a_1 v_1 + a_2 v_2$

Def: spanning set.

we say that a set $T = \{ v_1, v_2, \dots, v_n \}$ is a spanning set for a vector space V if $V = \text{span}(T)$.

Def: A basis of a v.s. V is a spanning set that is linearly independent.



Def: The dimension of a vectors space V is the number of elements of a basis for V.

If T is a basis for V : $\dim(V) = |T|$

Notation: Vectors $\begin{matrix} | & | & & | \\ p_1, & p_2, & \dots, & p_n \\ | & | & & | \end{matrix}$ where each $p_i \in \mathbb{R}^m$

Form $A = \begin{bmatrix} | & | & & | \\ p_1 & p_2 & \cdots & p_n \\ | & | & & | \end{bmatrix}_{m \times n}$

linear combination

$$\underline{x} = c_1 p_1 + c_2 p_2 + \dots + c_m p_m$$

$$= \underbrace{[\, p_1 \ p_2 \ \cdots \ p_n \,]}_{A} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

$= \quad A \subseteq \qquad \subseteq$

Def: Let V be a vector space. Then a real-valued function

$$\|\cdot\| : V \longrightarrow \mathbb{R}$$

is a <u>norm</u> if it satisfies the following:

N1: $\|x\| \geq 0 \quad \forall\, x \in V$

N2: $\|x\| = 0$ iff $x = 0$

N3: $\|\alpha x\| = |\alpha| \cdot \|x\|, \quad \forall\, \alpha \in \mathbb{R},\ x \in V$

N4: $\|x+y\| \leq \|x\| + \|y\|$ (triangle inequality)

For a metric space that is a v.s.

$$d(x,y) = \|x-y\|$$

$$\|x\| = d(x, 0)$$

In $\mathbb{R}^d$:

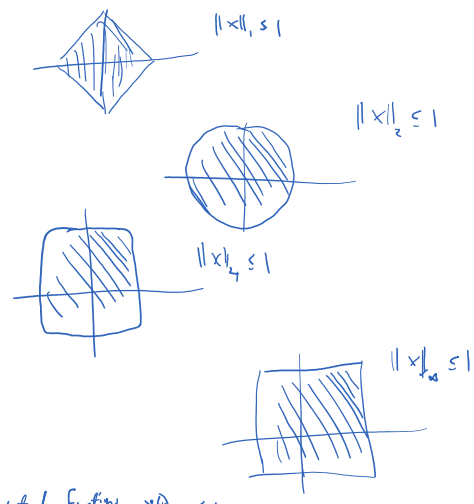
<u>Ex:</u>

$$\|x\|_1 = \sum_{i=1}^{d} |x_i| \qquad \ell_1$$

$$\|x\|_2 = \left(\sum_{i=1}^{d} |x_i|^2 \right)^{1/2} \qquad \ell_2$$

$$\|x\|_p = \left[\sum_{i=1}^{d} |x_i|^p \right]^{1/p} \qquad \ell_p$$

$$\|x\|_\infty = \max_{1 \leq i \leq d} |x_i| \qquad \ell_\infty$$



Ex: V = a space of functions.

$$\|x(t)\|_1 = \int_a^b |x(t)|\, dt$$

L_1: the set of functions $x(t)$ s.t. $\|x(t)\|_1 < \infty$

$$\|x(t)\|_2 = \left(\int_a^b |x(t)|^2\, dt \right)^{1/2}$$

L_2: the set of functions $x(t)$ s.t. $\|x(t)\|_2 < \infty$

$L_2[a,b]$

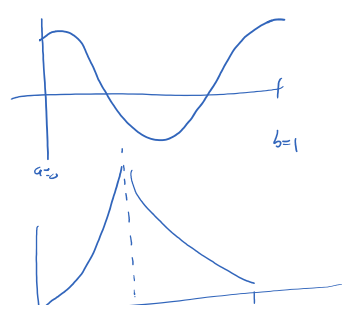
$L_2(-\infty, \infty)$.

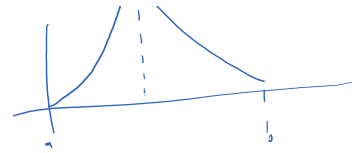
$$\|x(t)\|_p = \left(\int_a^b |x(t)|^p\, dt \right)^{1/p}$$

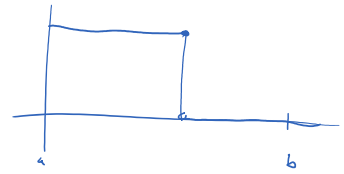
$$\|x(t)\|_\infty = \sup_{t \in [a,b]} |x(t)|.$$

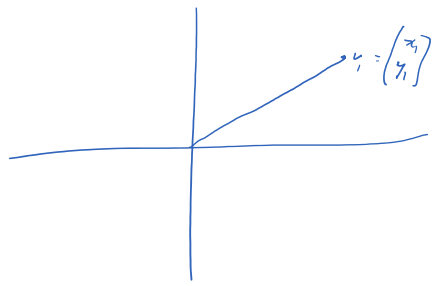
Q: $C[a,b]$

$L_2(a,b)$









$\| v_1 \|_2 = \sqrt{x_1^2 + y_1^2}$

Inner Product spaces

two slots

Def: Let V be a v.s. A function $\langle \, , \, \rangle : V \times V \longrightarrow \mathbb{R}$ is an <u>inner product</u> if it satisfies the following:

IP1: $\langle x, y \rangle = \langle y, x \rangle$ (real) $\qquad \langle x, y \rangle = \langle y, x \rangle^*$ (complex)

IP2: $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$ $\qquad \langle x, \alpha y \rangle = \alpha \langle x, y \rangle$ (real)

α : scalar. $\qquad \langle x, \alpha y \rangle = \alpha^* \langle x, y \rangle$ (complex)

IP3: $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$

IP4: $\langle x, x \rangle > 0$ if $x \neq 0$

$\langle x, x \rangle = 0$ iff $x = 0$

Ex: $\underline{x}, \underline{y} \in \mathbb{R}^n$

$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \qquad \underline{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$\langle \underline{x}, \underline{y} \rangle = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$$
$$= \sum_{i=1}^{n} x_i y_i$$
$$= \underline{x}^T \underline{y} = \underline{y}^T \underline{x}.$$

Ex: $\underline{x}, \underline{y} \in \mathbb{C}^n$

$$\langle \underline{x}, \underline{y} \rangle = \sum_{i=1}^{n} x_i y_i^*$$
$$= \underline{y}^H \underline{x}.$$

Ex: $L_2[0,1]$ <u>Real</u>

$x(t), y(t) \in L_2[0,1]$

$$\langle x(t), y(t) \rangle = \int_0^1 x(t) y(t)\, dt$$

$L_2[0,1]$ complex

$$\langle x(t), y(t) \rangle = \int_0^1 x(t)\, y^*(t)\, dt$$

Ex: $A \in \mathbb{R}^{m \times n}$

$B \in \mathbb{R}^{m \times n}$

$$\langle A, B \rangle = tr(B^T A)$$

$A \in \mathbb{C}^{m \times n}$

$B \in \mathbb{C}^{m \times n}$

$$\langle A, B \rangle = tr(B^H A).$$

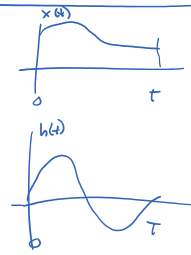
<u>Induced Norms</u>

$$\|x\|_2 = [\langle x, x \rangle]^{1/2}$$

Ex: $L_2(-\infty, \infty)$

$$\langle x(t), e^{j\omega t} \rangle = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, dt$$

Ex:



$$y(T) = x(t) * h(t)\Big|_{t=T}$$

$$= \int_0^T x(\tau)\, h(T-\tau)\, d\tau$$

Introduce: $g(\tau) = h(T-\tau)$

$$y(T) = \int_0^T x(\tau)\, g(\tau)\, d\tau$$

$$= \langle x(t), g(t) \rangle$$